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# Palaeomagnetic signature of slow post-orogenic cooling of the north-east Highlands of Scotland recorded in the Newer Gabbros of Aberdeenshire

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**Summary.** Consideration of previous work and new data show a systematic distribution of the directions of high coercivity and high blocking temperature magnetization in the Newer Gabbros of Aberdeenshire. The site-mean directions show a non-Fisherian distribution with the corresponding pole positions distributed along a published early Palaeozoic apparent polar wander path for Great Britain. The mean of the distribution closely agrees with the mean direction previously reported and widely quoted. There is a close correlation of the site-mean directions with the position of the sampling sites with respect to the contours ('chrontours') of the K–Ar cooling age. Sites where stable NRM has SE declinations and shallow (+20°) inclinations have the oldest chrontour ages (~468 Myr). The mean palaeomagnetic direction progressively shifts and the sites with the youngest chrontour ages (~428 Myr) have SW declinations with moderately steeper (+45°) inclination. The relative age of magnetization, determined from the position of the corresponding pole on the apparent polar wander path, is consistent with the relative chrontour age. The gabbros have therefore recorded the shift of the north-east Highlands with respect to the palaeomagnetic pole during the slow cooling of the terrain. The slow cooling is presumably related to uplift and erosion, and the resulting downward migration of isothermal surfaces. A previously published suggestion that all magnetizations with southerly declinations and moderately downward inclination are the resultant of normal and reversed magnetizations with shallow inclination is shown to be incorrect.

## Introduction

It is well known that apparent polar wander paths for the various continental blocks are a consequence of lateral plate motion relative to the poles: the palaeomagnetic directions vary systematically in rocks magnetized in successive geologic ages. The palaeomagnetic imprint of vertical motions of the lithosphere such as uplift and the accompanying erosion of orogenic belts is less well known. Studies of Precambrian terrains such as the Grenville province in Canada (McWilliams & Dunlop 1978), Greenland shield (Morgan 1976) and

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Limpopo belt in Africa (Morgan & Briden 1981) indicate that palaeomagnetic directions may have a systematic geographic distribution associated with the cooling history. In the particular case of the Grenville province, McWilliams & Dunlop (1978) have shown that the mean palaeomagnetic directions are correlated with potassium–argon cooling ages.

The cooling age pattern of the Palaeozoic metamorphic terrain in north-east Scotland, presumably the direct result of uplift and erosion, is quite well known. Dewey & Pankhurst (1970) reviewed the then-available K–Ar ages and drew lines of equal apparent age ('chrontours') for the whole region. It is the aim of this study to show that the slow cooling imparted a characteristic palaeomagnetic signature to the Newer Gabbros of Aberdeenshire. These rocks do not record a single mean palaeomagnetic direction, but rather a non-Fisherian distribution of site-mean directions that correspond with a segment of the published British apparent polar wander path for the Palaeozoic. The site-mean directions of high blocking temperature magnetizations carried primarily by magnetite are closely related to the potassium–argon 'chrontour' ages. To show this, evidence is drawn from previously published data and from new data presented in this study.

## Previous palaeomagnetic investigations

Blundell & Read (1958) measured the natural remanent magnetization (NRM) of samples collected from the major areas of outcrop of the Aberdeenshire gabbros to determine if the gabbro bodies have been rotated during late Caledonian folding. They did not subject the samples to demagnetization experiments but stability was tested by storage. Broadly similar though rather scattered directions of NRM were found in different bodies and they concluded that the gabbro had not been rotated since formation. Stewart & Johnson (1960) questioned this conclusion, discussing field evidence for rotation and asserted that the scatter of NRM directions might be a consequence of deformation. They also considered the possibility that the gabbro bodies may have been folded after crystallization, but before magnetization.

Sallomy & Piper (1973) completed an extensive survey of outcrops from the major areas of exposure. They carried out pilot alternating field demagnetization experiments on at least one specimen from each site to determine an optimum demagnetization field using the stability index of Briden (1972). At least one specimen from each site was thermally demagnetized, but site-mean directions reported by them are predominantly from single treatment alternating field demagnetization at the optimum field for each site. Sallomy & Piper (1973) concluded that the magnetization post-dates tilting with the possible exception of the Insch mass as the mean direction from this body was anomalous (SE declination and shallow inclination) compared to the other gabbro bodies. They noted in particular that the mean direction from the Boganclogh mass was very different from the Insch mass. This presented a problem as the Boganclogh mass is an extension of the Insch mass, the link between them being buried beneath a veneer of Old Red Sandstone. With this reservation, they computed the total mean direction of magnetization for the Newer Gabbros and an often-quoted early Ordovician pole position.

Carmichael & Storetvedt (1981) sought to refute the suggestion that this mean direction of magnetization is truly representative of the palaeomagnetic record of the gabbro. They pointed out that the demagnetization techniques employed by Sallomy & Piper (1973) may not have been sufficiently rigorous fully to reveal the potentially multivectorial nature of the magnetization of these rocks. On the basis of analysis of 12 independently oriented samples, they claim that the mean direction reported by Sallomy & Piper (1973) is the resultant of normal and reversed magnetization with shallow inclination. This argument is further investigated and rejected in a subsequent section of this paper.

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## Further analysis of the work of Sallomy & Piper (1973)

In an exercise similar to that carried out by McWilliams & Dunlop (1978), the sampling sites listed in table 1 of Sallomy & Piper (1973) were plotted on a map with the K–Ar biotite chrontours. The chrontours are shown on the geologic map in Fig. 1 (after Fettes & Chesher 1977). The ages are recalculated using the decay constant given by Steiger & Jager (1977). The sites may be classified into three roughly equal groups according to chrontour age for the calculation of mean directions and associated Fisher statistics (Table 1) from the site-mean palaeomagnetic directions after AF and thermal demagnetization given by Sallomy & Piper. Examination of Table 1 shows that the estimate of Fisher's precision parameter, $k$, of each group is larger than that of the data set as a whole. The cones of 95 per cent confidence overlap for each group, but the means shift systematically westward with younger chrontour age in a manner consistent with the published British apparent polar wander path (Briden & Duff 1981).

**Table 1.** Mean palaeomagnetic directions of subpopulations of mean site directions from Sallomy & Piper (1973).

| Category | $N$ | $\bar{D}$ | $\bar{I}$ | $k$ | $\alpha_{95}$ |
|---|---|---|---|---|---|
| $\geqslant$ 468 Myr | 12 | 153° | 33° | 14.6 | 12° |
| < 468 Myr and $\geqslant$ 443 Myr | 11 | 173 | 28 | 10.8 | 14 |
| < 443 Myr | 9 | 186 | 51 | 12.9 | 15 |
| Overall | 32 | 168 | 34 | 9.7 | 9 |

Category = position of sampling sites with respect to K–Ar biotite chrontours; $N$ = number of sites from which mean direction is computed; $\bar{D}$ = mean declination °E of N; $\bar{I}$ = mean inclination, positive down; $k$ = estimate of Fisher (1953) precision parameter; $\alpha_{95}$ = apical half-angle of cone of 95 per cent confidence.

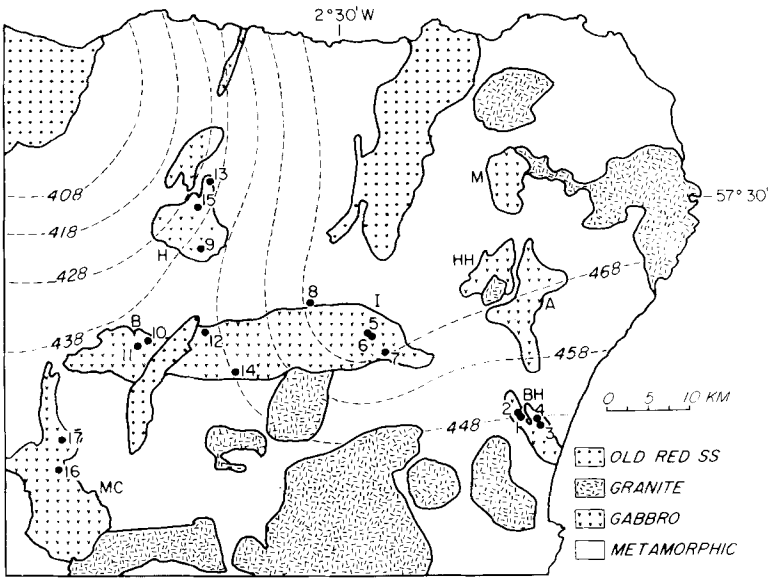



**Figure 1.** Map of the north-east Scottish Highlands (after Fettes & Chesher 1977) showing the generalized geology, palaeomagnetic sampling sites, and the biotite K–Ar apparent age chrontours calculated using the decay constant in Steiger & Jager (1977). The letters denote the individual gabbro masses: H – Huntly mass, MC – Morven Cabrach mass, B – Boganclogh mass, I – Insch mass, M – Maud mass, HH – Haddo House mass, A – Arnage mass, BH – Belhelvie mass. The numbers denote the sampling site numbers in Table 2.

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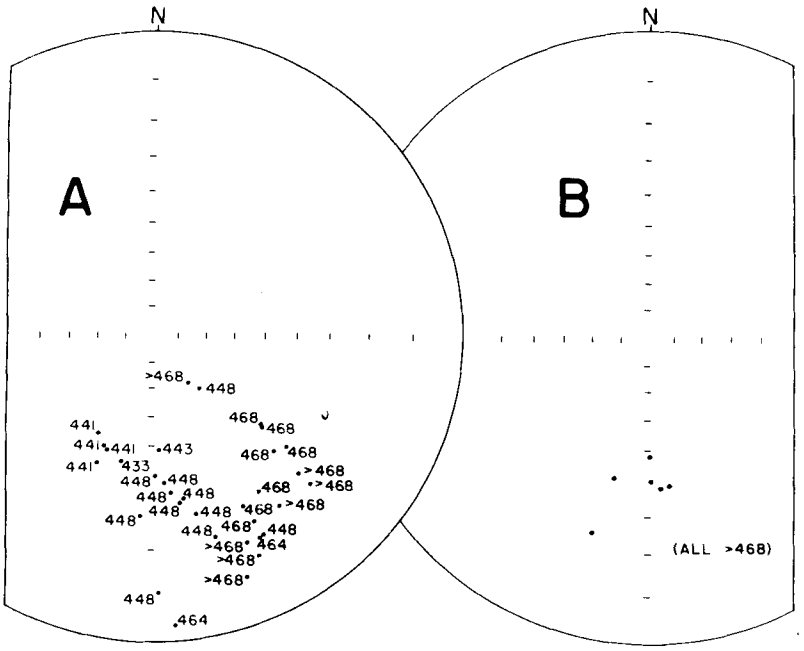



**Figure 2.** (a) Equal angle projection of the directions of components isolated from the original thermal demagnetization data used by Sallomy & Piper (1973) by the weighted principals component analysis algorithm of Kent *et al.* (1983). The numbers denote the chrontour age of the sampling site. The directions in (b) are from the Maud and Arnage masses, from sites with chrontour ages of 468 Myr or more.

The original stepwise thermal demagnetization data generated by Sallomy & Piper (1973) were re-analysed using the weighted principal component analysis algorithm of Kent, Briden & Mardia (1983). The best fitting components were found for demagnetization steps between 300 and 600°C. Fig. 2(a) is an equal area projection of the directions of these components with the associated chrontour age of the sampling site.

The oldest chrontour ages correspond to magnetization with SE declination and inclination approximately 20°. The youngest ages correspond to magnetization with SW declination and inclination about 45°, and the intermediate steps are well represented. The exception to this pattern are the results from the Maud and Arnage masses (shown in Fig. 2b). Although the chrontour age for these masses is greater than 468 Myr the directions of magnetization are most similar to the sites with chrontour ages of 448 Myr.

Without resorting to detailed statistical arguments, the results of these exercises were sufficiently intriguing to justify further investigation to test the possibility that the palaeomagnetic record of the Newer Gabbros is intimately associated with the protracted cooling history of the north-east Highlands, applying more detailed and rigorous demagnetization techniques.

## New palaeomagnetic data from the Newer Gabbross

Fig. 1 is a generalized geological map showing the location of the new sampling sites with respect to the K–Ar biotite chrontours. Site 8 is located in the metamorphic aureole of the Insch mass and all other sites are in igneous rocks. Most of the sites were located from the national grid coordinates given by Sallomy & Piper (1973). Between five and 10 cores per site were collected using a portable coring apparatus. The orientations of the cores were

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measured *in situ* using a clinometer, magnetic compass and, when possible, a solar compass. The cores were sliced into cylindrical specimens 2.54 cm wide and 2.3 cm high.

Magnetization was measured using a Digico complete results spinner magnetometer (Molyneux 1971) and the British prototype CCL cryogenic magnetometer. Laboratory treatment was adapated to the requirements of each site according to the material available and to the stability of magnetization which was monitored throughout the demagnetization procedure. Thermal demagnetization was emphasized to complement the detailed alternating field experiments carried out by Sallomy & Piper (1973) in their pilot studies which discriminated between low and higher coercivity magnetization components, which they considered secondary and primary respectively. Thermal demagnetization was carried out using a large furnace in which the external field is annulled by orthogonal coils with feedback control (McElhinny, Luck & Edwards 1971; McClelland-Brown 1980) in a nitrogen atmosphere. Alternating field demagnetization was done using a tumbler device with peak fields of 200 mT (McElhinny 1966).

At demagnetization temperatures above 600°C, viscous remanence acquired during the transfer of the specimen from the demagnetization device to the magnetometer posed problems for measurement, although the samples were transferred directly from the

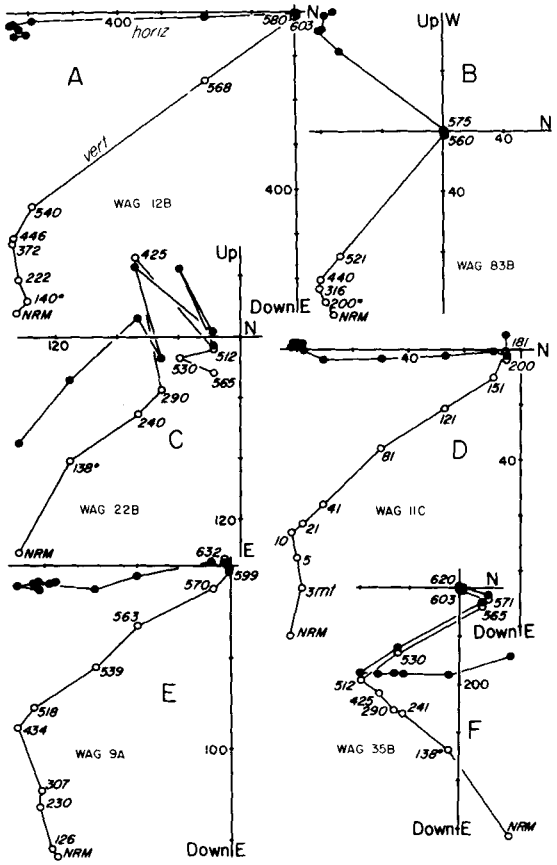



**Figure 3.** Orthogonal projections (Zijderveld 1967) of demagnetization results from current study. Open circles denote projection on the vertical plane. Closed circles denote projection on the horizontal plane. Demagnetization treatment is given either in degrees Centrigrate or millitesla (mT). The units of the axes are $10^{-3}$ A m$^{-1}$.



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demagnetization device to a mu-metal box and carried to the magnetometer. Using the cryogenic magnetometer, decay of the viscous magnetization could be directly observed and results were recorded when the output was stable for several minutes.

The demagnetization results were interpreted using orthogonal projections (Zijderveld 1967). The diagrams were generated during the demagnetization procedure and each sequence was deemed complete only after the endpoints reached the origin of the diagram. Each diagram was examined for linear demagnetization trajectories, i.e. three or more colinear points on both projections. Components so identified were isolated by the subtraction of the vectors defined by the two sets of points that best defined the trajectory. These interpretations were checked using the weighted principal component analysis algorithm of Kent *et al.* (1983) which fits straight lines and planes to a three-dimensional distribution of points taking into account the errors of measurement.

**Table 2.** Mean directions of characteristic NRM, statistics and poles.

| Site No. | National Grid Ref. | (Chrontour Age Myr) | Treatment | N | $N_s$ | $\bar{D}$ | $\bar{I}$ | k | $\alpha_{95}$ | Palaeomagnetic pole lat | Palaeomagnetic pole long | dp | dm |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | NJ921191 | (448) | thermal | 5 | (5) | 177° | 37° | 88 | 8° | -12° | 0° | | |
| | | | a.f. | 2 | (2) | 173 | 36 | 233 | 16 | -13 | 4 | | |
| | | | combined | 7 | (7) | 176 | 36 | 115 | 6 | -12 | 2 | 4.0° | 7.0° |
| 2 | NJ919194 | (448) | thermal | 5 | (5) | 182 | 38 | 33 | 13 | -12 | 356 | | |
| | | | a.f. | 3 | (3) | 168 | 40 | 50 | 18 | -9 | 8 | | |
| | | | combined | 8 | (8) | 177 | 39 | 36 | 9 | -11 | 1 | 6.4 | 10.7 |
| 3 | NJ944180 | (448) | thermal | 5 | (5) | 204 | 46 | 15 | 20 | -3 | 336 | | |
| | | | a.f. | 3 | (3) | 213 | 27 | 22 | 27 | -13 | 325 | | |
| | | | combined | 8 | (8) | 209 | 40 | 17 | 14 | -6 | 331 | 10.1 | 16.9 |
| 4 | NJ942187 | (448) | thermal | 4 | (5) | 181 | 31 | 128 | 8 | -16 | 357 | 5.0 | 8.9 |
| | | | a.f. | 0 | (2) | | | | | | | | |
| 5 | NJ737292 | (>468) | thermal | 4 | (4) | 150 | 27 | 80 | 10 | -14 | 27 | | |
| | | | a.f. | 3 | (3) | 155 | 14 | 66 | 15 | -22 | 24 | | |
| | | | combined | 7 | (7) | 152 | 21 | 52 | 8 | -18 | 26 | 4.4 | 8.4 |
| 6 | NJ738289 | (>468) | thermal | 5 | (5) | 142 | 20 | 64 | 10 | -16 | 36 | | |
| | | | a.f. | 1 | (1) | 136 | 2 | - | - | - | - | | |
| | | | combined | 6 | (6) | 141 | 17 | 47 | 10 | -17 | 38 | 5.3 | 10.3 |
| 7 | NJ758267 | (468) | thermal | 5 | (5) | 152 | 36 | 78 | 9 | -9 | 24 | | |
| | | | a.f. | 3 | (3) | 149 | 27 | 37 | 20 | -14 | 28 | | |
| | | | combined | 8 | (8) | 151 | 32 | 54 | 8 | -11 | 26 | 5.1 | 9.0 |
| 8 | NJ668327 | (468) | thermal | 4 | (5) | 142 | 21 | 147 | 8 | -15 | 36 | | |
| | | | a.f. | 2 | (3) | 138 | 15 | 46 | 38 | -16 | 42 | | |
| | | | combined | 6 | (8) | 141 | 19 | 96 | 7 | -16 | 38 | 3.8 | 7.3 |
| 9 | NJ397536 | (446) | thermal | 5 | (6) | 177 | 58 | 12 | 23 | 6 | 0 | 24.9 | 33.9 |
| | | | a.f. | 0 | (2) | | | | | | | | |
| 10 | NJ474284 | (441) | thermal | 5 | (5) | 210 | 42 | 253 | 5 | -4 | 330 | | |
| | | | a.f. | 1 | (1) | 211 | 45 | - | - | - | - | | |
| | | | combined | 6 | (6) | 210 | 43 | 303 | 4 | -4 | 330 | 3.1 | 5.0 |

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Table 2 (continued)

| Site No. | National Grid Ref. | (Chrontour Age Myr) | Treatment | N | $N_s$ | $\bar{D}$ | $\bar{I}$ | k | $\alpha_{95}$ | Palaeomagnetic pole lat | Palaeomagnetic pole long | dp | dm |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 11 | NJ461278 | (441) | thermal | 6 | (6) | 214 | 39 | 146 | 6 | -5 | 325 | | |
| | | | a.f. | 3 | (3) | 204 | 38 | 88 | 13 | -8 | 334 | | |
| | | | combined | 9 | (9) | 211 | 39 | 107 | 5 | -6 | 328 | 3.6 | 6.0 |
| 12 | NJ542294 | (446) | thermal | 4 | (5) | 162 | 44 | 18 | 22 | -6 | 14 | 17.2 | 27.5 |
| | | | a.f. | 0 | (3) | | | | | | | | |
| 13 | NJ549478 | (433) | thermal | 5 | (5) | 202 | 40 | 27 | 15 | -8 | 336 | | |
| | | | a.f. | 2 | (2) | 212 | 44 | 25 | 52 | -3 | 328 | | |
| | | | combined | 7 | (7) | 205 | 41 | 29 | 11 | -6 | 334 | 8.1 | 13.4 |
| 14 | NJ579247 | NRM's were less than $5 \times 10^{-3}$ $Am^{-1}$. Thermal and alternating field demagnetization resulted random vector variations except for one specimen which showed a linear trajectory with a direction of 163/-24 and blocking temperatures up to 570°C. | | | | | | | | | | | |
| 15 | NJ533444 | 95% of the NRM was removed by either 350° thermal demagnetization or by 10 mT maximum alternating field. At 10 mT treatment, the remanence direction measurement was not repeatable. | | | | | | | | | | | |
| 16 | NJ365127 | Apparently random vector variations during thermal and alternating field magnetization. | | | | | | | | | | | |
| 17 | NJ370165 | | | | | | | | | | | | |

'Treatment' denotes the results derived from thermal or alternating field demagnetization, or the mean of results of the two methods; $N$ = number of results used to calculate the mean direction; $N_s$ = total number of specimens measured; $\bar{D}$ = mean declination °E of N; $\bar{I}$ = mean inclination (positive downward); $k$ = estimate of Fisher (1953) precision parameter; $\alpha_{95}$ = apical half-angle of the cone of 95 per cent confidence; lat, long are latitude (positive N) and longitude (positive E) of the corresponding palaeomagnetic pole position; *dp, dm* = semi-axes of the oval of 95 per cent confidence associated with the pole position (Creer, Irving & Runcorn 1957).

Fig. 3 shows representative orthogonal projections of the magnetization vector during treatment. Fig. 3(a, b) illustrates characteristic behaviour, which may be summarized as the removal of a low blocking temperature component with steep, downward inclination between room temperature and 400°C to reveal a high blocking temperature component generally removed between 450 and 600°C, with southerly declination and low to moderate positive inclination. Several sites exhibited behaviour characterized by Fig. 3(c) which shows a distinct low blocking temperature component, but a highly erratic demagnetization trajectory during stepwise demagnetization at higher temperatures. No interpretation for this behaviour is offered here and the magnetization of these rocks is not deemed representative of the palaeomagnetic field. Fig. 3(d) shows the characteristic response to alternating field demagnetization. A steep inclination component is removed in a peak field of 10 mT to reveal a high coercivity component which decays uniformly in higher demagnetization fields until 180 mT at which point the remanence moves in a random, non-reproducible manner. Fig. 3(e) shows somewhat more erratic directional changes during thermal demagnetization, but the algorithm of Kent *et al.* (1983) shows that when the accuracy of individual measurements is considered, a single component is recognized between 434 and 570°C. An exceptional result from site 5 is shown in Fig. 3(f) where three components of magnetization are clearly revealed. The two components which are typically seen in the majority of sites are removed by demagnetization steps up to 570° to show a third component with NE declination and shallow inclination with a maximum blocking temperature of 620°C. This behaviour was observed only during thermal demagnetization of three specimens from the single site.

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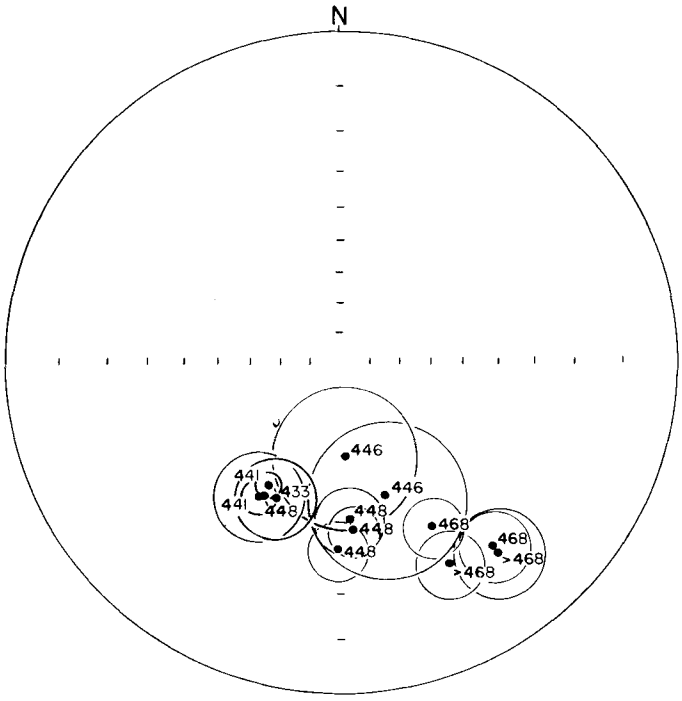



**Figure 4.** Equal angle projection of site-mean directions of magnetization with associated circles of 95 per cent confidence and chrontour ages of the sampling sites. The ages only denote the position of the site with respect to the chrontours and do not necessarily imply an absolute age of magnetization (see final section of text).

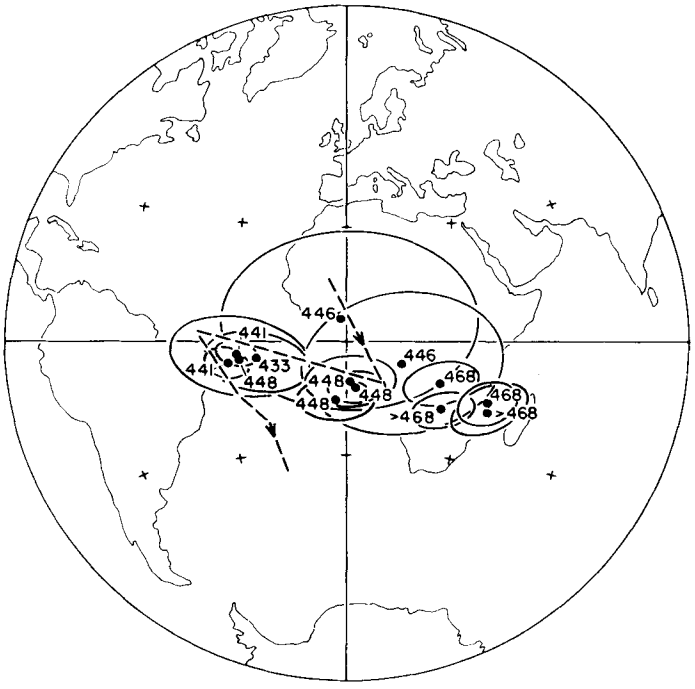



**Figure 5.** Site-mean palaeomagnetic poles with associated ovals of 95 per cent confidence and chrontour ages plotted on the British apparent polar wander path of Briden & Duff (1981).

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The palaeomagnetic record of the contact-altered thermal aureole rocks (site 8) is similar to that characteristic of the gabbro except the high-temperature magnetization persisted up to 670°C.

In Table 2 are listed the site mean directions with associated Fisher statistics and pole positions calculated from the directions of magnetizations with coercivities greater than 10 mT and blocking temperatures between 400 and 600°C for the gabbro, and 500 and 670°C for the aureole. Sites which yielded uninterpretable results or did not have magnetizations with blocking temperatures above 350°C are also listed in this table with a description of the behaviour.

Fig. 4 is a stereographic projection of the site-mean directions with associated cones of 95 per cent confidence. The ages indicate the position of the sampling site with respect to the K–Ar biotite chrontours. As was indicated by the re-analysis of the data of Sallomy & Piper (1973), there is a very clear association between the mean direction and chrontour age as the sites with the oldest chrontour ages have SE declinations and shallow inclinations and the sites with younger chrontour ages have SW declinations and moderately down inclinations. Fig. 5 shows the site-mean pole positions with ovals of 95 per cent confidence labelled with the associated chrontour age. The poles are distributed along the apparent polar wander path of Briden & Duff (1981) with the chrontour ages younging in the same sense as the polar wander path.

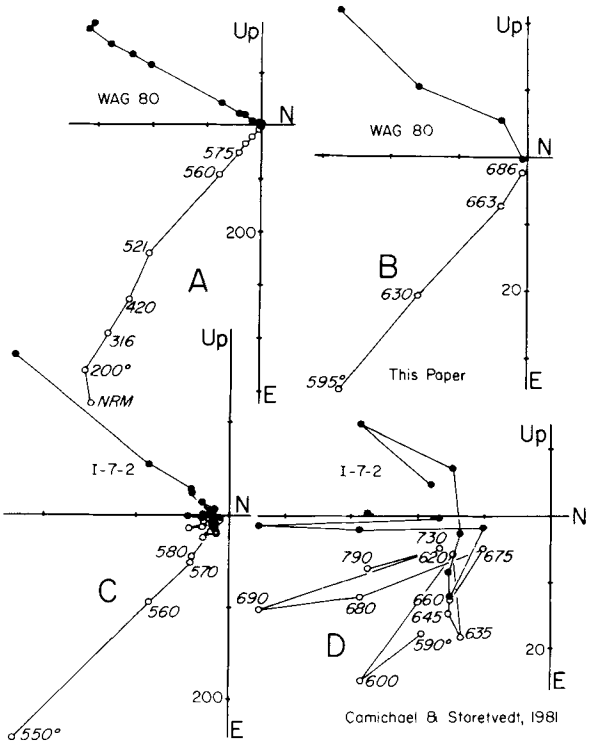



**Figure 6.** Orthogonal projections comparing results from the current study (a, b) to results published by Carmichael & Storetvedt (1981) (c, d) for the same sampling site (NJ 474284). (b) and (d) are expanded projections of (a) and (c) respectively showing detailed behaviour near the origin. The convention of plotting and the units are the same as Fig. 3.

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**Discussion of Carmichael & Storetvedt (1981)**

Carmichael & Storetvedt (1981) claim that the magnetization of the gabbro is multi-component and the major components are both normal and reversed, with shallow inclinations and declinations directed NNW–SSE and NNE–SSW. They contend that the mean direction calculated by Sallomy & Piper (1973) is a vector sum of unresolved components with these orientations, that remanence observed after AF demagnetization or thermal demagnetization below 600°C is most likely a resultant not representative of the direction of the inducing field, that the correct direction is revealed by thermal demagnetization at temperatures above 600°C, and this may persist up to 790°C. These conclusions are clearly at variance with results presented in this paper.

One site (our site 10, Carmichael & Storetvedt samples I7, I8) is common to the two studies and the conflict can be assessed directly. Fig. 6(a) is an orthogonal projection of thermal demagnetization of a sample from this site with Fig. 6(b) an expanded projection showing the last four steps of the sequence. Fig. 6(c, d) shows the corresponding plots for sample I7 of Carmichael & Storetvedt (1981) reconstructed from their figs 3 and 6. Up to temperatures of 590°C, a similar response is observed with a uniform decay of the remanence toward the origin at demagnetization temperatures ahove 500°. Above 600°C (Fig. 6b, d), there is marked difference. Fig. 6(d) must be viewed with considerable caution because it was reconstructed from a small published stereographic projection and intensity curve; while directions are probably retrieved with accuracy better than 3°, relative intensity could well be in error by as much as 15 per cent. Nevertheless we can see in Fig. 6(d) the alternatively increasing and decreasing intensity in the shallow southern direction which Carmichael & Storetvedt claimed to represent the primary magnetization. Hence it is probably justifiable to conclude from Fig. 6(d) that the results of Carmichael & Storetvedt (1981) are erratic above 600°C and that if their chosen high blocking temperature component is real at least four more components with blocking temperatures above 600°C must be explained! Failing that, their data above 600°C must be dismissed as 'noise'. By contrast, our data show that the high blocking temperature remanence (Fig. 6b) has the same direction as the lower blocking temperature remanence found by us (Fig. 6a) and by Carmichael & Storetvedt (Fig. 6c). Moreover, our measurements at 686°C have intensity

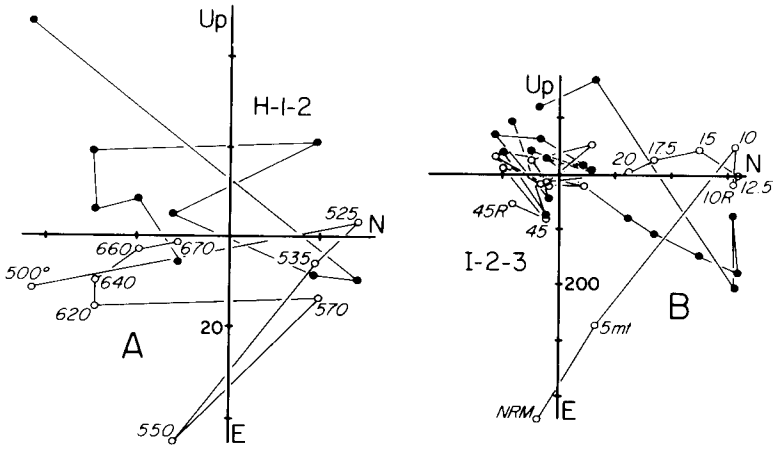



**Figure 7.** Orthogonal projections constructed from data of Carmichael & Storetvedt (1981). The convention of plotting and the units are as in Fig. 3.

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0.4 per cent of the NRM with the magnetization showing pronounced viscous behaviour and the direction is unlikely to be significant. The extrapolated linear trajectory misses the origin by less than 0.2 per cent of total NRM and this is a measure of the noise in our measurements. This example seems to indicate that the noise in Carmichael & Storetvedt's data is very much greater than ours, and that their claim of shallow inclination magnetization with blocking temperatures above 600°C and up to 790°C is unfounded at least for this particular site. Also, the precise linearity of the demagnetization trajectory between 500 and 600°C in both our work and that of Carmichael & Storetvedt argues against the remanence with blocking temperatures in that range being composite as claimed by those authors; our data show that this is indeed the only stable magnetization characteristic of these rocks.

Inspection of other data of Carmichael & Storetvedt (1981) suggests that these same critical conclusions apply. For example, their specimen H-1.2 (Fig. 7a) is claimed by them to carry a shallow south-westerly component with blocking temperatures 620–660°C and to illustrate 'the interplay between normal and reversed magnetizations with shallow inclination' below 600°C. Their AF demagnetization of specimen I-2.3 is said to show a 'very poorly defined shallow inclination component with a SSW declination at treatment above 25 mT'; Fig. 7(b) illustrates their demagnetization results for this specimen and requires no further comment.

## Discussion and conclusions

The elongate distribution of site-mean directions of magnetization found in our study could in principle arise from post-cooling structural movements, apparent polar wander during slow cooling or a combination of these processes.

The relationship of the gabbro to the structural history has been the subject of some discussion as reviewed by Fettes (1970). Read (1923) suggested the gabbro was post-tectonic as the metamorphic aureole about the Insch mass apparently overprints or destroys the cleavage in the Macduff Slates. Shackleton (in discussion of Read & Farquhar 1956) maintained that the present distribution of the gabbro was due to the folding of a single basic sheet in the Boyndie Syncline and Buchan Anticline. Stewart & Johnson (1960) supported this view and suggested that the cleavage in the Macduff Slates, axial planar to the D3 Boyndie Syncline was not destroyed by the metamorphic aureole, but had never developed near the Insch mass.

Fettes (1970) observed two generations of cleavage in the aureole of the Insch mass. The cleavage he assigned to $S_1$ is progressively destroyed towards the mass. In the outer zone of the aureole, andalusite and cordierite overgrow $S_1$. A second cleavage assigned to $S_3$, disrupts some of the recrystallized textures suggesting that $S_3$ post-dates the aureole. However, Fettes (1970) also observed that cordierite continued to crystallize after $S_3$, implying the rocks were still hot during $S_3$ development. From other observations on the distribution of the regional metamorphic isograds, Fettes concluded that the gabbro was intruded during the main regional Buchan metamorphism, and from the pattern of $S_1$ cleavage attitute around the Insch–Boganclogh mass, this body acted as a rigid strut during D3 movements.

The elongate distribution of mean-site directions is evident within the Insch–Boganclogh mass alone with supporting evidence from the Belhelvie and Huntly masses. If this distribution were due entirely to differential structural rotation of originally identical directions, this implies a difference of attitude between the eastern and western parts of the Insch–Boganclogh mass as great as 100° if rotations were about horizontal axes. If the eastern part were horizontal (i.e. the palaeomagnetic direction is truly representative of the inducing field direction) then the western part of the mass would be overturned with respect to the east.

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It is concluded that structural displacements alone cannot be invoked to explain the distribution of the mean-site directions in the Newer Gabbros. Instead, we argue that the correlation of site-mean directions with the chrontour age is due to change of the geomagnetic field during prolonged slow cooling when the rocks had already acquired essentially their present attitude. It would be very fortuitous if randomizing structural displacements would act in a manner to disperse the palaeomagnetic directions so that the corresponding poles lie along the published British apparent polar wander path (Briden & Duff 1981) in the proper time sequence. It would be equally fortuitous for the Huntly and Belhelvie masses to now lie at the same attitude as the parts of the Insch–Boganclogh mass with the same chrontour age.

The slow cooling of the terrain was presumably a consequence of uplift and erosion. At each site, the magnetization would have been recorded during the time interval spanned by the downward passage of isothermal surfaces corresponding to the blocking temperature distribution of the magnetic grains. The components of magnetization identified as characteristic of the gabbros have blocking temperatures approximately between 450 and 600°C with the lower temperature a minimum estimate. The trajectories of the projection of the decay of the magnetization on the orthogonal diagrams are not curved. Apparent polar wander does not appear to be recorded in single sites. The time required for a site to acquire the characteristic magnetization appears to be short compared to time-scales of apparent polar wander. However, the time for the entire terrain to acquire its palaeomagnetic signature appears to be long compared to time-scale of apparent polar wander, i.e. of the order of 30–40 Myr.

There is a hint in the Sallomy & Piper (1973) data set from the Maud and Arnage masses that the published cooling age pattern does not agree with the palaeomagnetic signature (Fig. 2b), perhaps requiring a revision of the chrontour pattern, and this is the subject of further work now in hand.

A surprising feature of these data is the small proportion of normal polarity magnetization. Only in site 5 is a magnetization of possible normal polarity identified in the current study and Carmichael & Storetvedt (1981) may have found another example (10–25 mT steps in Fig. 7b). If the time required to magnetize the terrain was *c.* 40 Myr, the palaeomagnetic field almost certainly changed polarity at least several times during this interval. The polarity history of the palaeomagnetic field in the Ordovician (438–505 Myr, Harland *et al.* 1982) remains to be documented. However, information on the polarity bias can be gathered from the *Geophysical Journal* palaeomagnetic pole lists (McElhinny 1969, 1970, 1972b, a; McElhinny & Cowley 1977, 1978, 1980). Of the 65 palaeomagnetic pole positions of probable Ordovician age given in lists X to XVI, 32 were derived from data of 100 per cent reversed polarity, 19 from data of mixed polarity and 14 from data of 100 per cent normal polarity. This simple exercise would seem to indicate that the Ordovician palaeomagnetic field was biased toward reversed polarity. If the time-scale required for a site in the gabbro to acquire its characteristic magnetization was long compared to the duration of constant polarity chrons, the terrain would have recorded a reversed polarity bias.

The absolute ages of magnetization are difficult to infer. If the magnetizations are TRM then chrontour ages must be regarded as minimum ages because of the difference of blocking temperature between argon in biotite and muscovite and magnetization in small grains of magnetite. The ages of magnetization would be older than the chrontour ages by the amount of time required for the terrain to cool from ~ 500°C to perhaps ~ 350°C.

However, TRM origin is by no means proven, the apparent polar wander path is far from perfectly calibrated, and the 'chrontour' pattern is certainly over-generalized. All these have to be attended to before the relative and absolute time span of magnetization of the Aberdeenshire Gabbros can be resolved.

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